AI-Powered Space-Air-Ground Networks for 6G

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Abstract

This paper presents an extensive study on the integration of artificial intelligence (AI) in Space-Air-Ground Integrated Networks (SAGIN) to improve operational efficiency, flexibility, and reliability in next-generation communication systems. The traditional SAGIN model has limitations in handling dynamic network conditions, task management, resource management, routing, and communication reliability in diverse space, air, and ground layers. To address these limitations in traditional SAGIN models, AI-based techniques like Deep Reinforcement Learning (DRL), Large Language Models (LLMs), and multi-agent techniques are proposed for smart decision-making and autonomous management in SAGINs. This paper proposes an AI-based SAGIN model that enables cross-layer optimization and adaptation in communication systems. The proposed model is evaluated using analytical models and simulation techniques to show significant improvements in resource management, trajectory optimization, and communication management compared to traditional models. This paper will also discuss some of the major challenges in integrating AI in SAGINs, including high computation complexities, data types, security risks, and potential research directions in this field.

Introduction

Conventional approaches to managing Space-Air-Ground Integrated Networks (SAGIN) face significant limitations in adapting to dynamic network conditions, which result in inefficient resource allocation, suboptimal trajectory planning, and unreliable communication across SAGIN. This creates significant challenges for maintaining network stability, ensuring seamless connectivity, and supporting real-time decision-making in next-generation communication systems [1]-[20]. To address these issues, recent research has explored the integration of artificial intelligence (AI) techniques and modules into the SAGIN architecture. Currently, AI-based solutions are being used to improve spectrum management, trajectory optimization, and communication handoff processes, resulting in more adaptable and automated network control. Approaches such as reinforcement learning, deep learning, large language models, and multi-agent systems demonstrate strong potential to improve performance and automate operations. Despite these advancements, existing studies have focused their research on specific applications and AI techniques, lacking a comprehensive perspective on their effectiveness across diverse problems in SAGIN. As a result, it is difficult to assess the impact that AI can provide to SAGIN environments. Therefore, a comprehensive review of the literature is required to assess current findings and identify research gaps. This section will focus on analyzing and organizing existing studies on key functional areas and methodologies of AI applied in SAGIN.

Background

The SAGIN is a multi-tier framework designed to provide the wide and seamless connectivity needed for 6G networks. It combines a space layer made up of Low Earth Orbit (LEO) and Geostationary Earth Orbit (GEO) satellites for global coverage, an air layer with Unmanned Aerial Vehicles (UAVs) and High-Altitude Platform Stations (HAPS) for flexible on-demand support, and a ground layer made up of traditional terrestrial cellular networks. By connecting these layers, SAGIN can extend high-speed communication to remote, maritime, and aerial areas that are difficult for standard infrastructure to reach. However, managing SAGIN is challenging because the network topology changes constantly and each layer has very different resource constraints. Fast-moving satellites and changing UAV flight paths can lead to frequent link disruptions, while differences in power, computing capacity, and mobility make control difficult. This can create the problem of outdated information, where network states change faster than control signals can propagate.

To deal with these challenges, artificial intelligence is being used to make SAGIN more adaptive and autonomous. DRL is useful for solving dynamic optimization problems, such as trajectory planning and resource allocation, because it learns from interactions with the environment. LLMs and Agentic AI also support intent-based networking, where high-level service requirements can be turned into actions across the space, air, and ground layers with less manual control. In addition, Edge AI and Function-as-a-Service (FaaS) move computation closer to where data is generated, which helps reduce latency and improve reliability for time-sensitive 6G applications.

Literature Review: AI-Empowered SAGIN

The development of 6G networks relies on the integration of terrestrial, aerial, and space-based communication systems into a unified SAGIN. Unlike traditional terrestrial networks, SAGIN operates in dynamic and heterogeneous environments, where satellites, UAVs, and ground stations must coordinate under changing network conditions, mobility constraints, and environmental factors such as orbital movement, signal interference, and limited energy resources. Because of this complexity, traditional rule-based network management approaches are often too rigid to handle real-time decision-making and cross-layer device coordination. As a result, recent research has increasingly focused on artificial intelligence as a way to improve adaptability, efficiency, and automation in SAGIN systems.

This literature review focuses on four areas that are essential to AI-empowered SAGIN: spectrum and resource allocation, service-oriented network slicing and software-defined networking, trajectory optimization, and handoff and beamforming mechanisms. These areas represent the core operational challenges in SAGIN, as they address how network resources are distributed, how different services are supported, how mobile nodes are positioned, and how communication is maintained across moving platforms. Together, the studies reviewed in this section demonstrate how AI techniques such as reinforcement learning, graph neural networks, and agentic AI are being used to improve coordination and decision-making across space, air, and ground network layers.

Spectrum and Resource Allocation

Spectrum and resource management in Space-Air-Ground Integrated Networks is challenged by highly dynamic network conditions, limited spectrum availability, and conditional changes like orbital drift or atmospheric effects [1]. As a result, conventional SAGIN systems rely significantly on rule-based logic, which is incapable of adapting to constantly changing and unpredictable environments. Additionally, feeder links, which are communication links that connect satellites to the ground, suffer from severe signal loss because of atmospheric interference or long distances [2]. This creates signal degradation, which makes it challenging for SAGIN to transmit large volumes of data [1]. Furthermore, each network component has unique challenges to overcome. Satellites move fast but predictably, aerial vehicles such as drones are adaptable but energy-constrained, and ground terminals are stable but limited by their infrastructure [1]. These challenges make it difficult to manage resources like bandwidth, routing, and task management across a large network.

Subsequently, traditional SAGIN architecture is limited by spectrum shortage and the presence of numerous systems, such as UAVs, satellites, and terrestrial networks, in the same frequency bands [3]. Currently, SAGIN systems frequently share the same frequency bands, leading to network overlap, interference, signal congestion, and impaired communications when multiple devices attempt to transmit large volumes of data simultaneously. Furthermore, traditional SAGIN models struggle to balance prioritizing dependable connections across devices, resulting in service degradation during peak traffic periods [3].

Despite these challenges, artificial intelligence offers a solution that can address bottlenecks in spectrum management and resource allocation. To begin, Deep Reinforcement Learning modules are able to analyze historical and current data flowing through SAGIN systems. Instead of following a static rule-based protocol, DRL can learn from the data it's given and predict the best strategies to apply depending on the condition [1]. DRL, for example, can use historical data to determine which frequency bands are least congested at different times of day, automating channel allocation to UAVs, which reduces network interference and enables smooth communication [3]. Additionally, Large Language Models can also analyze historical usage data to forecast massive traffic sessions, automatically adjusting the distribution of network bandwidth based on the usage levels of SAGIN systems[5]. Furthermore, Graph Neural Networks provide intelligent routing, link prediction, and situational awareness within SAGIN [1]. GNNs can provide smart routing in global networks to quickly make decisions that will provide stability during seamless handoffs. GNNs can optimally identify paths for data to flow, which can minimize delays and reduce energy consumption significantly. Additionally, it can also help in load balancing between fast-moving satellites and process multiple model data to produce a graph displaying environmental understanding [1]. In uncertain and changing settings, GNN can analyze its surroundings and use various model data to self-adapt and distribute resources.

While AI can significantly automate and improve spectrum management and resource allocation, it also plays a major role in enabling service-oriented network slicing within SAGIN. This capability allows the network to partition physical resources into multiple virtual slices, each configured to meet the performance requirements of 6G applications, such as low-latency communication or high-traffic environments.

Service-Oriented Network Slicing and SDN

Building upon the AI resource allocation strategies previously discussed, the next step in SAGIN is the implementation of a service-oriented architecture. This involves using Software-Defined Networking (SDN) to dynamically slice the network, ensuring that the spectrum is used efficiently across diverse application domains.

To support the heterogeneous requirements of 6G, Cheng et al. propose a service-oriented SAGIN architecture centered on network slicing [4]. This approach allows the physical network infrastructure to be divided into multiple virtual networks that can be customized for specific use cases. As illustrated in Fig. 1, SAGIN combines satellite, aerial, and terrestrial devices while supporting different service-oriented slices for applications such as emergency response, automated driving, and smart city sensing. For example, certain slices can be configured with high-reliability and low-latency to support automated driving, while others can handle high-density environments such as smart city sensors or traffic management systems [4].

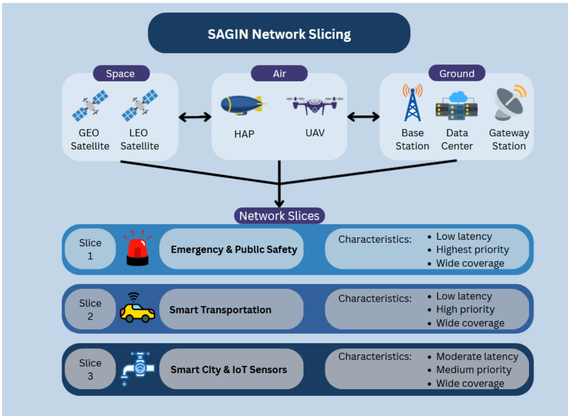

Figure 1: Overview of Space-Air-Ground Integrated Network (SAGIN) with example network slices.

This orchestration is enabled by SDN, which uses a centralized controller to manage data flow and configure network devices across different layers of the system [4]. The framework also incorporates cloud-edge synergy, where satellite, aerial, and terrestrial resources work together to provide both wide-area coverage and localized computing capabilities [4]. In this architecture, artificial intelligence, particularly DRL, can be used to automate the creation of network slices and dynamically allocate bandwidth based on real-time network demand [4].

However, a challenge in this model is the impact of propagation delays and the continuous movement of satellites. By the time information regarding the network status reaches the centralized controller, the data may already be outdated due to signal travel time and changes in satellite position. This ”stale” information makes it difficult for AI-driven controllers to maintain an accurate, real-time view of the global network, potentially leading to inefficient resource allocation or suboptimal network performance.

The issues of propagation delay and outdated network status show a disconnect between logical network orchestration and physical node mobility. To reduce these disruptions and maintain the integrity of service-oriented slices, the system must proactively manage node movement, which is addressed through Trajectory Optimization (TO).

Trajectory Optimization

In SAGIN, nodes are deployed across space, aerial, and terrestrial layers, all of which require intelligent and dynamic network management. In both the aerial and space layers, dynamic conditions cause coverage gaps, requiring nodes to be repositioned to maintain network connectivity. UAVs also have the problem of energy constraints due to their limited battery capacity, making energy-efficient TO a challenge. Due to these dynamic conditions, TO becomes a necessity to maintain network performance, energy efficiency, and stability. As a result, some researchers have proposed a combination of RL approaches to optimize UAV and HAPS trajectory and positioning [5]

In a study conducted by Abohashish et al. [5], their objective was to test and study a UAV-trajectory optimization (UAV-TO) scheme for up-link communication using load balancing to maximize energy efficiency, resulting in higher network resource utilization. Their results found that the proposed UAV-TO scheme utilizing RL significantly improved performance compared to traditional circular and linear schemes. Specifically, of the two RL approaches used in the scheme, Monte Carlo (MC) and dynamic programming (DP), MC proved to be more energy efficient across all situations, especially as flight duration increased, and had the highest peak within suburban areas.

In contrast, Kharchenko et al. [6] focused on testing packet loss through a simulated SAGIN channel using a Remotely Piloted Aircraft System (RPAS) equipped with AI. This is particularly relevant to TO, as UAV trajectory and positioning directly influence transmission distance and therefore signal strength, which in turn affects packet loss and overall reliability within SAGIN systems. Their results found that packet loss was heavily dependent upon transaction size (TS) and data transmission rates.

At low TS the system displayed minimal loss, but as the TS reached up to 100,000 bits, this caused excessive packet loss and eventual complete failure of the channel. The channel closure represents a practical consequence of what happens when the system has a lack of adequate load balancing.

These studies demonstrated that TO affects not only energy efficiency but also communication reliability and network performance. While Abohashish et al.[5] focused on energy efficiency using load balancing, Kharchenko et al.[6] focused on communication reliability, measuring packet loss through an AI-integrated SAGIN environment. These studies indicate that multiple network parameters must be considered when attempting to optimize trajectory in SAGIN networks.

While combinations of RL frameworks serve as the foundation of TO, AI is evolving towards the use of Agentic AI which employ the use of LLMs. This transition towards LLMs enables the shift from reactive protocols to a proactive approach in which the system can adapt and learn [7].

Dev et al. [7] proposed an architectural framework change that decouples the user and control planes, allowing AI to manage them independently. This enables specialized AI agents to independently manage routing policies and packet forwarding pathways. By using intent-based orchestration, the LLM-powered agents can execute an execution-feedback-adjust-monitoring loop to manage node positioning and resource allocation more efficiently than traditional methods.

By employing new technologies like Neural Radio Protocol Stacks (NRPS) that allow for dynamic protocol creation and which allow for functions to be computed on the network edge, this enables Agentic AI to execute intent based-networking with less resource use. Their results using Agentic AI in a V2X simulation demonstrated an improved packet delivery ratio of 79% to 95%, a reduction in retransmission rates from 17% to 5% and a reduction in latency from 50ms to 20ms.

Despite the promising results demonstrated by RL and agentic AI for TO, most of these studies were conducted in simulated environments with perfect conditions, where a single optimization was chosen. As a result, the performance results represent ideal optimizations rather than realistic deployment scenarios. In real SAGIN systems where conditions are highly dynamic, TO must be calculated while considering parameters such as energy consumption, communication reliability, and latency.

While TO provides a physical foundation for communication stability and energy efficiency, it alone is insufficient to ensure reliable communication in dynamic SAGIN environments. As nodes reposition, the rapid movement introduces propagation delays, which require additional mechanisms such as beamforming, MIMO, and seamless handoff strategies to ensure connection stability and signal quality.

Handoff, Beamforming, MIMO

Currently, traditional SAGIN satellite systems cannot provide coverage in all areas. As a result, they resort to beam hopping, which focuses wireless signals in specific directions rather than spreading them throughout the area. Moreover, with the high mobility of satellites and UAVs, beamforming struggles to steer itself and direct the signal to hotspots or specific users in a region. This demonstrates a major challenge within satellite systems because multiple satellites are attempting to provide coverage for multiple areas [1]. This becomes more problematic when deciding which area should receive the most coverage at specific time intervals.

Despite these existing challenges, multi-agent reinforcement learning (MADRL) can bring many satellites together, forming a team of AI agents who collaborate to make predictive decisions and deliver the greatest coverage[1]. MADRL can leverage the information gathered to automatically coordinate beam-hopping and beamforming decisions, addressing inefficiency and coverage difficulties in standard SAGIN architectures. To further mitigate these spatial constraints, the deployment of Reconfigurable Intelligent Surfaces (RIS) on aerial platforms establishes indirect line-of-sight links that bypass physical blockages in urban ”dead zones” [8]. These surfaces induce precise phase changes to reconfigure propagation, which, when integrated with Deep Learning-based Massive MIMO, significantly optimizes link quality and mitigates Doppler frequency offsets[9]. The transition toward Wireless Big AI Models (wBAIM) enables a unified scheduling paradigm that synchronizes user motion status across heterogeneous tiers, ensuring that seamless signal handoffs occur without service interruption[10]. Ultimately, utilizing machine learning to estimate precise channel state information with reduced overhead maintains robust connectivity and service continuity across the dynamic 6G topology[8].

Discussion

From reviewing a number of research papers pertaining to AI empowered SAGIN networks, a pattern of challenges and approaches to incorporating AI is evident. Traditional rule based protocols are too rigid for the highly adaptive nature of SAGIN networks. The use of DRL to manage data routing is discussed in a number of papers and how various methods can be optimized for accuracy and efficiency. A 2025 article by Liu et al. discusses the use of a GCN-Transformer hybrid model to optimize routing decisions based on network structure and traffic patterns. Hashima et al. examines how RL is used to optimize spectrum and power usage. The use of RL to manage limited resources and routing creates a more efficient network, as is also discussed by Wang et al.

The need for AI to manage complex SAGIN networks is well established, however, implementing and training these AI models requires careful consideration. The papers examined discuss various machine learning methodologies, each targeting different optimization goals. Given the difficulties of transferring or processing large amounts of data on satellites and other aerial devices, Split Learning (SL) is considered a viable solution; allowing satellites and ground stations to share the load of training the AI model [11]. Federated Learning (FL) allows the preservation of privacy since training occurs locally and only trained model data is uploaded to other devices, not raw data [12, 4]. Reinforcement Learning (RL) is the most prevalent technique discussed in the papers examined and excels in optimization and control [4]. A variation known as Multi Agent Deep Reinforcement Learning has been shown effective for decentralized routing and beamforming on satellites [13].

The trustworthiness and reliability of AI remain concerns that are frequently discussed. Dev et al. describes AI as a black-box with limited transparency and compliance [7]. The need for combining rigid frameworks with AI is critical because the inherent opacity of complex models, such as deep neural networks, makes them susceptible to backdoor and adversarial attacks. Such attacks can trick the system into producing false or malicious results [14]. This lack of transparency creates a significant barrier for safety-critical applications, such as autonomous vehicles, where an auditable trail may be required for regulatory standards [7]. To solve this problem, researchers have explored Explainable AI (XAI) frameworks like SliceOps, which is designed to provide transparent reasoning behind its decisions [15]. Dev et al. suggest further research into hybrid symbolic-AI, where neural networks are combined with human-curated knowledge bases, creating an auditable decision trail [7].

Energy consumption and sustainability is another area of interest in regards to SAGIN networks. Although some papers briefly discuss sustainability, the majority of discussion pertains to energy consumption of aerial devices. Satellite processing capabilities and energy supply is limited due to their reliance on solar panels and the thermal cooling constraints of space [16]. Developing more energy efficient ways to process and route data is of significant interest for satellite based networks.

As previously mentioned, Split Learning can be used to reduce processing requirements onboard satellites by shifting part of the load to ground stations. The reliance on satellite processing can also be reduced through the use of transparent payloads, or satellites that reflect signals from ground stations without regenerating the signal [8]. Dev et al. briefly mentions the sustainability concerns around AI and its growing computational demands, requiring significant energy consumption [7]. Resource intensive AI solutions must be balanced against the benefits they provide.

Research Gaps and Proposed Future Research

Energy consumption of AI remains a major limiting factor in its use with non-terrestrial networks. The research examined shows that AI has many potential uses and advantages in satellite networks, however, energy constraints onboard satellites remain a challenge. The benefits of AI must be carefully balanced against the drawbacks of increased energy use. Determining what amount of onboard AI processing is most beneficial is a research topic that remains difficult to answer, but contributes towards the success of SAGIN networks. Additional research into reducing energy consumption of satellite AI would prove beneficial in increasing SAGIN network viability.

The research papers examined briefly discuss or make limited mention of airborne platform stations (APSs), such as HAPSs, drones, and balloons. These types of devices offer advantages over LEO satellites, such as lower cost, stationary positioning, lower latency, and minimal path loss [13]. Although many researchers agree that APSs have a place in SAGIN networks, few real-world, specifically large-scale, tests have been conducted. Hashima et al. discusses simulated testing of various UAVs, but does not conduct any real-world tests. Kharchenko et al. also performed simulated testing of UAVs, explaining that testing in real networks is difficult because of AI compatibility with existing hardware [6]. Although real-world tests of non-satellite aerial devices may be difficult and costly, they are a necessary step in proving their viability in SAGIN networks.

The research examined highlights that while AI can significantly improve SAGIN network performance, its adoption in satellite systems remains constrained by onboard energy limitations. Future research should focus on reducing the power consumption of AI and evaluating the balance between its drawbacks and benefits. Additionally, more research involving real-world and large-scale testing of APSs is needed beyond simulated tests. Testing of UAVs, HAPSs, and balloon platforms in realistic environments and with realistic network conditions would provide critical validation of APS viability in SAGIN networks.

Conclusion

SAGIN's innovative aspect is the integration of AI techniques, including multi-agent reinforcement learning, intent-based large language models, and emerging models such as Wireless Big AI Models, into Space-Air-Ground Integrated Networks, enabling autonomous, intelligent, and adaptive network management. Traditional SAGIN architectures rely on rule-based logic and centralized control, which are inefficient when handling dynamic conditions such as orbital drift, atmospheric interference, and energy constraints.

MADRL enables satellites, UAVs, and terrestrial networks to collaborate as agents, making real-time decisions for beamforming, beam-hopping, and network coverage optimization. Furthermore, MADRL can boost network efficiency in high-traffic areas. Furthermore, LLM agents can help with intent-based recognition by making predictions about routing, resource allocation, and SAGIN positioning. GNNs can help with predictive routing and load balancing by optimizing data flow between space, air, and ground layers while reducing energy usage.

Service-Oriented network slicing divides a physical network into multiple virtual networks, each optimized for specific requirements such as reliability, low latency, or high traffic volume. When used with wBAIM, the SAGIN system may synchronize slice scheduling across many layers, resulting in seamless connectivity and effective resource allocation. SAGIN can adapt proactively, manage resources across the three layers, and ensure stable connectivity in dynamic 6G settings by using several AI-driven models, architectures, and methods.

Glossary

Term	Definition
Deep Reinforcement Learning (DRL)	A machine learning model that learns from historical data and interactions to make predictions and guide future decisions.
Edge AI	AI that runs directly on local devices, enabling real-time processing for tasks like security, authentication, and privacy in networks such as 6G.
Geostationary Earth Orbit (GEO)	Satellite orbital path with an altitude of approximately 35,786 kms and a relative motion that remains stationary to Earth.
Graph Neural Network (GNN)	A neural network that learns from connected data by understanding how nodes and their relationships influence each other.
High-Altitude Platform Station (HAPS)	Semi-permanent, aerial devices such as airships and balloons that typically operate at altitudes between 17 - 22 km.
Large Language Models (LLMs)	AI models trained on massive datasets that can understand and generate human-like text for tasks such as writing, summarizing, and question answering.
Low Earth Orbit (LEO)	Satellite orbital path with an altitude of 200 – 2,000 km and a relative velocity of roughly 7.6 km/s.
Software-Defined Networking (SDN)	A networking approach where a centralized software controller manages and controls how data moves through the network.
Space-Air-Ground Integrated Network (SAGIN)	Heterogeneous network architecture combining ground-based networks, air-based networks, and space-based networks.

Term	Definition
Unmanned Aerial Vehicle (UAV)	Also referred to as a drone, a UAV is a highly mobile aerial device that can operate autonomously or via remote pilot control.

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